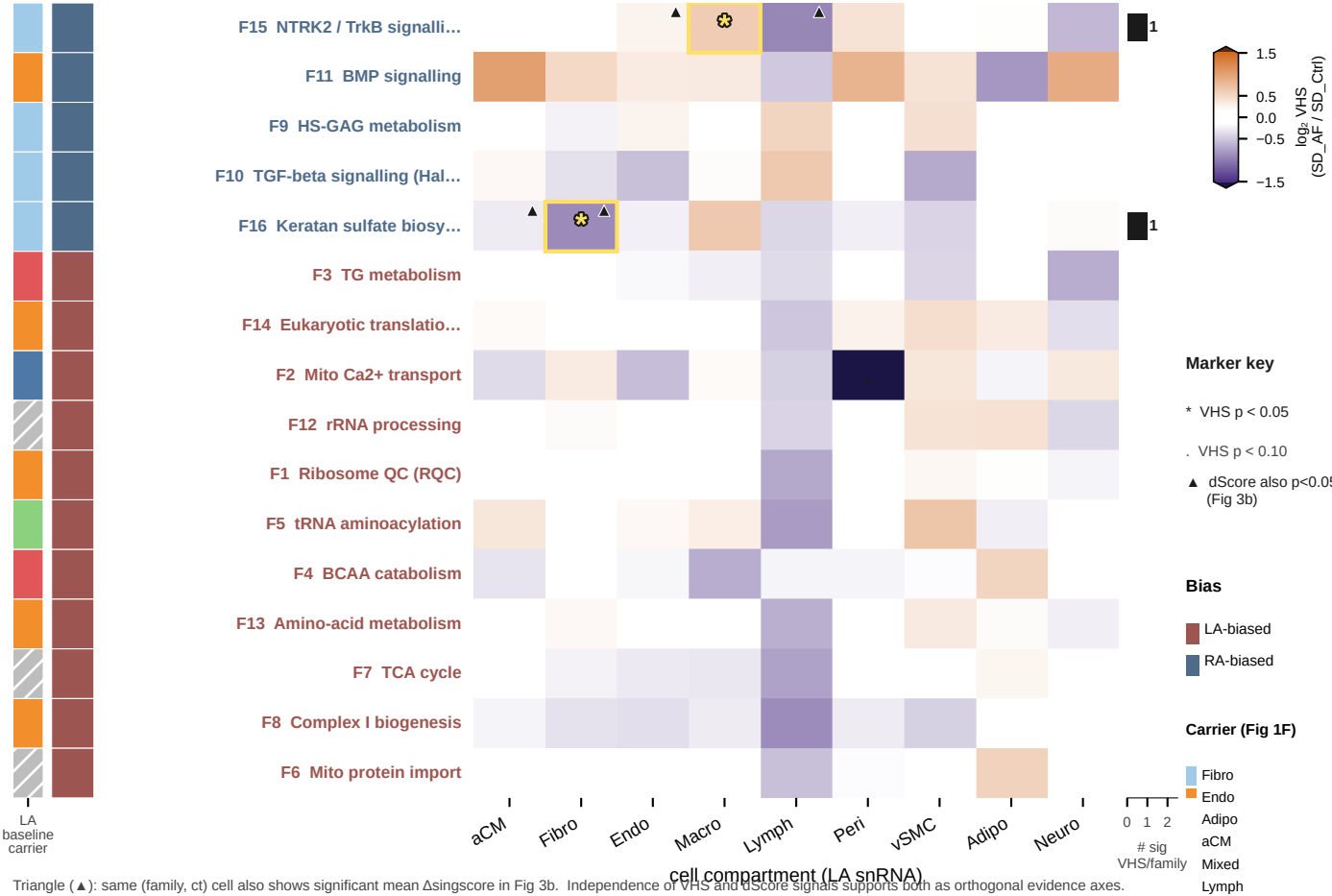
Fig S3 (VHS) Variance-Heterogeneity Score: hidden sub-population signal beyond mean Δsingscore



Triangle (▲): same (family, ct) cell also shows significant mean Δsingscore in Fig 3b. Independence of VHS and dScore signals supports both as orthogonal evidence axes.

Purple (log_2 VHS < 0): AF cohort more uniform than Ctrl → cell-intrinsic homogenisation or composition lock-in.

Orange (log_2 VHS > 0): AF cohort more dispersed than Ctrl → candidate sub-population emergence in AF (hidden by mean dScore).

VHS = SD_AF(family singscore) / SD_Ctrl(family singscore) per (family x compartment) on GSE255612 LA snRNA pseudobulk.